Reg. No. | | | | | | | | | | | [18BSPH02]

SECOND SEMESTER B.Tech EXAMINATION, JUNE–2019

PHYSICS

Time: 3 Hours **Maximum Marks:70**

Instructions:
- i. Answer **Five** full Questions, each question carries 14 marks.
- ii. **Question no.1 and 2 are compulsory.**
- iii. Missing data may be suitably assumed.

List of Physical constants:

- i. Velocity of light: $c = 3\times10^{8}$ m/s
- ii. Planck's constant: $h = 6.625\times10^{-34}$ J.S
- iii. Mass of neutron, $m_n = 1.675\times10^{-27}$kg
- iv. Charge of electron: $e = 1.6\times10^{-19}$ C
- v. Mass of electron: $m_e = 9.1\times10^{-31}$ kg
- vi. Boltzmann constant, $k = 1.38\times10^{-23}$ J/k
- vii. Permittivity of free space: $\varepsilon_0 = 8.854\times10^{-12}$ F/m
- viii. Permeability of free space, $\mu_0 = 4\pi\times10^{-7}$H/m

1.	a.	Derive an expression for internal field in one dimensional solid dielectrics.	07
	b.	Distinguish between soft and hard magnetic materials.	04
	c.	Find the polarization produced in a crystal by an electric field of strength 500 V/mm if it has a dielectric constant of 6.	03
2.	a.	Set up time independent Schrodinger's wave equation in one-dimension.	06
	b.	Derive the relation between group velocity and phase velocity.	04
	c.	A particle of mass $0.5\text{MeV}/c^2$ has a kinetic energy 100eV. Find its deBroglie wavelength, where c is the velocity of light.	04
3.	a.	Discuss the variation of Fermi factor with energy and temperature.	07
	b.	List the failures of classical free electron theory.	03
	c.	Find the temperature at which there is 1% probability that a state with energy 0.5ev above the Fermi energy is occupied.	04

OR

4.	a.	Derive an expression for the density of states for conduction electrons for unit volume of metal.	09
	b.	Discuss how quantum free electron theory is successful over classical free electron theory.	05

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5. a. Derive an expression for inter-planar spacing in terms of Miller indices. 06

b. Define: 04

i. Coordination number

ii. Space lattice

iii. Bravais lattice

iv. Unit cell

c. If the lattice constant of a cubic crystal is $3x10^{-10}$ m, find the interplanar spacing between (111) planes. 04

OR

6. a. Explain the crystal structure of diamond with neat diagram. 06

b. Define atomic packing factor. Calculate the atomic packing factor for bcc structure. 04

c. Find the Miller indices of a set of parallel planes which make intercepts in the ratio 3a:4b on X and Y axes and are parallel to Z-axis. a, b and c are the primitive vectors of the lattice. 04

7. a. Explain the construction and working of a carbon dioxide LASER with neat diagram. 06

b Define spontaneous and stimulated emission with suitable diagrams. 04

c A laser operating at 632.8 nm emits $3.182x10^{16}$ photons per second. Calculate the output power of the laser if the input power is 100 watt. Also find the percentage power converted into coherent light energy 04

OR

8. a. Derive an expression for numerical aperture in an optical fiber. 06

b. Explain single mode fiber & graded index multimode fiber with suitable diagrams. 04

c. Calculate the V-number for a fiber of core diameter 40□m & with refractive indices of 1.55 and 1.5 respectively for core and cladding when the wavelength of the propagating wave is 1400 nm. Also calculate the number of modes that the fiber can support for propagation. Assume that the fiber is in air. 04